B. Sc. 2nd Semester (Honours) Examination, 2025

Mathematics

Course ID: 22114 **Course Code: SH/MTH/203/GE-2**

Algebra

[NEW CBCS]

Time: 2 Hours. **Full Marks: 40**

The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.

Symbols and notations have their usual meanings.

Answer all the questions.

UNIT-I

1. **Answer *any five* of the following questions.** $2 \times 5 = 10$

a) From a bi-quadratic equation with rational coefficients two of whose roots are $2i \pm 1$.

b) If $z_1 \& z_2$ be two complex number, prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.

c) Let $a \sim b$ mean that $a \equiv b \pmod n$, where a, b in $\mathbb{Z}$ and n is fixed positive integer. Show that $\sim$ is an equivalence relation.

d) Determine the eigen values of the matrix $A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 6 \end{pmatrix}$.

e) Obtain the column rank of the matrix $A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{pmatrix}$

f) Find Ker T for the linear mapping $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + 2y + 3z, 3x + 2y + z, x + y + z)$.

[Turn Over]

g) If $\alpha, \beta, \gamma, \delta$ are roots of the equation $x^4 + px^3 + qx^2 + rx + s = 0$, then find the value of $\sum \frac{\alpha}{\beta}$.

h) Suppose $T: \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation with $T(1,-1,0) = (2,1), T(0,1,-1) = (-1,3), T(0,1,0) = (0,1)$. Find $T(-1,1,2)$.

UNIT-II

2. Answer *any four* of the following questions. $\quad 5 \times 4 = 20$

a) If $\alpha_1, \alpha_2, \dots\dots\dots, \alpha_n$ be n distinct roots of $x^n + nax + b = 0$, then prove that $(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \dots\dots\dots\dots (\alpha_1 - \alpha_n) = n(\alpha_1^{n-1} + a)$.

b) Let X be a non-empty set and $P(X)$ be the power set of X. Let us define a relation $' \leq '$ on $P(X)$ by $A \leq B$ means $'A$ is subset of $B' \forall A, B \in P(X)$. Establish that $(P(X), \leq)$ is poset.

c) If a, b, c are all positive real numbers such that $abc = \lambda^3$, prove that $(1+a)(1+b)(1+c) \geq (1+\lambda)^3$. When the equality occurs? 4+1

d) for what values of λ and μ do the system of equations have
(i) no solution (ii) unique solution (iii) infinite solution?
$x + y + z = 6$
$x + 2y + 4z = 10$
$x + 2y + \lambda z = \mu.$

e) State the Fundamental Theorem of Arithmetic. Find integers u, v satisfying $30u + 72v = 12$.

f) Solve the equation $16x^4 - 64x^3 + 56x^2 + 16x - 15 = 0$ whose roots are in arithmetic progression.

UNIT-III

3. Answer *any one* of the following questions. **$10 \times 1 = 10$**

a) (i) Let $\mathbb{Z}$ be the set of all integers and a relation R is defined on $\mathbb{Z}$ by aRb means $a - b$ is divisible by 7. Prove that R is an equivalence relation.
(ii) Solve the bi-quadratic equation $x^4 - 6x^2 + 16x - 15 = 0$ by Ferrari's method.
(iii) Use the principle of induction to prove that $1 + 3 + 5 + \cdots + (2n - 1) = n^2 \ \forall n \in \mathbb{N}$.

b) (i) Find out the matrix corresponding to the linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = \frac{1}{2}(4x + 4y + 2z, -x + y + 3z)$ with respect to the ordered basis $\{(0,1,1), (1,0,1), (1,1,0)\}$ of $\mathbb{R}^3$ and $\{(1,0), (1,1)\}$ of $\mathbb{R}^2$.
(ii) Using Sturm's functions, find the nature of real roots of the equation $x^5 - 5x + 5 = 0$.
(iii) Find the inverse of the matrix $\begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$ using the Cayley-Hamilton theorem. 4+3+3
